

MAT1100 Homework 1

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Question 6

- (a) *Proof.* Since $G/Z(G)$ is cyclic it is generated by a single coset. Let z be a representative of this coset. Then all cosets of $Z(G)$ in G are of the form $z^a Z(G)$ for some $a \in \mathbb{Z}$.

Now consider $x, y \in G$ they each lie in some coset of $Z(G)$, therefore there exist $a, b \in \mathbb{Z}$ such that $x \in z^a Z(G)$ and $y \in z^b Z(G)$. Therefore there exist $g, h \in Z(G)$ such that $x = z^a g$ and $y = z^b h$. Now compute

$$\begin{aligned} xy &= z^a g z^b h \\ &\text{since } g \in Z(G): \\ &= z^a z^b g h \\ &\text{since } h \in Z(G): \\ &= z^{a+b} h g \\ &= z^b z^a h g \\ &\text{since } h \in Z(G): \\ &= z^b h z^a g \\ &= yx \end{aligned}$$

hence G is abelian. □

- (b) *Proof.* Let p be prime and G a group of order p^2 . Let $G \curvearrowright G$ by conjugation, then let $c(x_1), \dots, c(x_k)$ be the distinct non-trivial conjugacy classes of G , hence $|c(x_i)| > 1$. By orbit-stabilizer:

$$|c(x_i)| = |G|/|G_{x_i}| = p^2/|G_{x_i}|$$

since $|c(x_i)| > 1$, $|G_{x_i}| < p^2$ which implies that $|G_{x_i}| \in \{1, p\}$. Now if $|G_{x_i}| = 1$ then $|c(x_i)| = |G|$ which implies $e \in c(x_i)$. Since conjugacy classes are disjoint or equal this forces $c(x_i) = c(e)$ which means $|c(x_i)| = 1$ contradicting $|c(x_i)| = |G| > 1$.

Thus $|G_{x_i}| = p$ and therefore $|c(x_i)| = p$. Now writing the class equation of G

$$|G| = |Z(G)| + \sum_{i=1}^k |c(x_i)|$$

then mod p , $|G| = 0$ and $|c(x_i)| = 0$ therefore this implies

$$0 = |Z(G)| \pmod{p}$$

now together with Lagrange's theorem this implies $|Z(G)| \in \{p, p^2\}$.

Case 1: $|Z(G)| = p^2$ then $Z(G) = G$ and then clearly G is abelian

Case 2: $|Z(G)| = p$ then $|G/Z(G)| = p$. Picking any non identity element $g \in G/Z(G)$ (which exists as $p > 1$) we must have $|g| = p$ by Lagrange's theorem and hence $G/Z(G) = \langle g \rangle$ meaning it is cyclic. Then by part (a), G is abelian. $\square$

- (c) *Proof.* Consider $G = D_8$. It has a cyclic normal subgroup $R = \langle r \rangle$, this is clear as $D_8 = \langle r, s \rangle$ and for any r^a we have

$$sr^a s^{-1} = r^{-a} s s^{-1} = r^{-a} \in R$$

then $|D_8/R| = 2$ which implies it is cyclic. However D_8 is not abelian as

$$sr = r^{-1}s$$

this is not a contradiction as $Z(G) = \langle e, r^2 \rangle$ which is a proper subgroup of R . And moreover $D_8/Z(G)$ is isomorphic to the Klein 4-group which is not cyclic. $\square$